

Optimal Production Planning in a Stochastic Manufacturing System with Long-Run Average Cost¹

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Abstract. This paper is concerned with the optimal production planning in a dynamic stochastic manufacturing system consisting of a single machine that is failure prone and facing a constant demand. The objective is to choose the rate of production over time in order to minimize the long-run average cost of production and surplus. The analysis proceeds with a study of the corresponding problem with a discounted cost. It is shown using the vanishing discount approach that the Hamilton–Jacobi–Bellman equation for the average cost problem has a solution giving rise to the minimal average cost and the so-called potential function. The result helps in establishing a verification theorem. Finally, the optimal control policy is specified in terms of the potential function.

Key Words. Production planning, stochastic dynamic programming, vanishing discount approach, optimal control, long-run average cost.

1. Introduction

There is a considerable literature devoted to optimal production planning in continuous-time manufacturing systems consisting of failure-prone

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machines with discounted cost criteria; see, e.g., Sethi and Zhang (Ref. 1) for references and details. Not much is available, however, in connection with the long-run average cost criterion. The exceptions are the papers by Bielecki and Kumar (Ref. 2) and by Ghosh, Arapostathis, and Marcus (Ref. 3).

Bielecki and Kumar (Ref. 2) deal with a single machine with two states (up and down), single product problem with linear holding, and backlog costs. Because of the simple structure of their problem, they were able to obtain an explicit solution for the problem, and thus verify the optimality of the resulting policy. When their problem is generalized to convex costs, explicit solutions are no longer possible. One then needs to develop appropriate dynamic programming equations, existence of their solutions, and verification theorems for optimality. This, however, has not been done in the literature. As a result, some generalizations of the Bielecki-Kumar problem such as those by Sharifnia (Ref. 4), Liberopoulos and Hu (Ref. 5), and Gershwin (Ref. 6) are only heuristic in nature.

Ghosh, Arapostathis, and Marcus (Ref. 3) introduced a nondegenerate diffusion term into the dynamics of the manufacturing system. This allowed them to rigorously analyze the resulting average cost minimization problem. They were able to obtain a sufficiently smooth solution of the dynamic programming equation in the presence of the diffusion process, and thus verify the solution to be the value function for their problem.

In this paper, we deal with a single or parallel machine manufacturing system with convex cost of holding and backlogging and without diffusion. As no explicit solution is available for the problem, we cannot extend the analysis of Bielecki and Kumar to treat the problem. We are also not able to use the results of Ghosh, Arapostathis, and Marcus (Ref. 3), since they assume a nondegenerate diffusion. Instead, we develop a new much-needed rigorous analysis to address the problem. We realize that the problem considered is not as general as some of the problems analyzed in the discounted cost framework. However, the discounted-cost production planning problems have been extensively studied over the last ten years [see Sethi and Zhang (Ref. 1)], whereas it has not been so for the average-cost problems. Our initial study is concerned with a simple model which is sufficiently general to make precise the heuristic treatments of several manufacturing system problems carried out in Refs. 4, 6 and others.

The plan of the paper is as follows. In Section 2, we introduce the problem and specify the required assumptions. Section 3 is devoted to the study of the value function associated with the corresponding control problem with a discounted cost. It establishes the results needed for the so-called vanishing discount approach often used for analysis of average cost minimization problems. In particular, we derive a key lemma showing that

one can go from any point in the state space to any other point in finite time. In Section 4, the Hamilton–Jacobi–Bellman (HJB) equation is specified for the average cost problem, and a verification theorem for optimality over the class of admissible controls is given. Moreover, by using the vanishing discount approach, it is shown that the HJB equation has a viscosity solution, which turns out in this case to be also a classical solution. The analysis helps us in Section 5 to specify the optimal control policy for the average cost problem under consideration. Section 6 provides a flowchart of the steps used in the vanishing discount approach and concludes the paper. Some required technical results are collected in the Appendix (Section 7).

2. Problem Formulation

We consider a single product manufacturing system with stochastic production capacity and constant demand for its production over time. In order to specify the model, let $x(t)$, $u(t)$, d denote the surplus level (state variable), the production rate (control variable), and the constant demand rate (parameter), respectively. We assume that $x(t) \in \mathbb{R} = (-\infty, \infty)$, $u(t) \in \mathbb{R}^+ = [0, \infty)$, $t \geq 0$, and d a positive constant. Surplus refers to inventory when $x(t) \geq 0$ and backlog when $x(t) < 0$. The system equation is

$$\dot{x}(t) = u(t) - d, \quad x(0) = x. \quad (1)$$

Let $\alpha(\cdot)$ denote a Markov process with a finite state space $\mathcal{M} = \{0, 1, 2, \dots, m\}$, with $\alpha(t)$ representing the maximum production capacity of the system at time t . The representation for $\mathcal{M}$ stands usually, but not necessarily, for the case of m identical machines, each with a unit capacity and having two states: up and down. This is not an essential assumption. In general, $\mathcal{M}$ could be any finite set of nonnegative numbers representing production capacities in the various states of the system.

Definition 2.1. A production control process $u(\cdot) = \{u(t), t \geq 0\}$ is admissible if:

- (i) $u(t) \in \mathcal{F}_t \equiv \sigma(\alpha(s), 0 \leq s \leq t)$;
- (ii) $0 \leq u(t) \leq \alpha(t)$, for all $t \geq 0$.

We denote by $\mathcal{A}(k)$ the collection of admissible controls with the initial condition $\alpha(0) = k$.

Definition 2.2. A function $w(x, k)$ defined on $\mathbb{R} \times \mathcal{M}$ is called an admissible feedback control, or simply a feedback control, if:

- (i) for any given initial surplus x and production capacity k , the equation

$$\dot{x}(t) = w(x(t), \alpha(t)) - d$$

has a unique solution;

- (ii) the control defined by $u(\cdot) = \{u(t) = w(x(t), \alpha(t)), t \geq 0\} \in \mathcal{A}(k)$.

With a slight abuse of notation, we simply call $w(x, k)$ a feedback control when no ambiguity arises.

Let $h(\cdot) : \mathbb{R} \mapsto \mathbb{R}^+$ and $c(\cdot) : [0, m] \mapsto \mathbb{R}^+$ denote the surplus (inventory/backlog) cost and the production cost, respectively. For any $u(\cdot) \in \mathcal{A}(k)$, define

$$J(x, k, u(\cdot)) = \limsup_{T \rightarrow \infty} (1/T) E \int_0^T (h(x(t)) + c(u(t))) dt, \quad (2)$$

where $x(\cdot)$ is the surplus process corresponding to the production process $u(\cdot)$. Our goal is to choose $u(\cdot) \in \mathcal{A}(k)$ so as to minimize the cost functional $J(x, k, u(\cdot))$.

We assume that the cost functions $h(\cdot)$, $c(\cdot)$ and the production capacity process $\alpha(\cdot)$ satisfy the following hypotheses:

- (A1) $h(\cdot)$ is a nonnegative convex function with $h(0) = 0$. There are positive constants C_1 , C_2 , and $\kappa_0 \geq 1$ such that

$$h(x) \geq C_1 |x|^{\kappa_0} - C_2, \quad x \in \mathbb{R};$$

moreover, there are constants $C'_1 > 0$ and $\kappa \geq \kappa_0$ such that

$$|h(x) - h(y)| \leq C'_1 (1 + |x|^{\kappa-1} + |y|^{\kappa-1}) |x - y|, \quad \forall x, y \in \mathbb{R}.$$

- (A2) $c(\cdot)$ is a nonnegative twice continuously differentiable function defined on $[0, m]$ with $c(0) = 0$. Moreover, for convenience in exposition, $c(\cdot)$ is either strictly convex or linear.

- (A3) $\alpha(\cdot)$ is a finite state Markov chain with generator Q , where $Q = (q_{ij})$, $i, j \in \mathcal{M}$, is an $(m+1) \times (m+1)$ matrix such that $q_{ij} \geq 0$, for $i \neq j$, and $q_{ii} = -\sum_{j \neq i} q_{ij}$. We assume that Q is strongly irreducible in the following sense: the equations

$$\nu Q = 0 \quad \text{and} \quad \sum_{i=0}^m \nu_i = 1$$

have a unique solution $v = (v_0, v_1, \dots, v_m)$, with $v_k > 0$, $k = 0, 1, \dots, m$. The vector v is called the equilibrium distribution vector of the Markov chain $\alpha(\cdot)$.

(A4) The average capacity $\bar{a} \equiv \sum_{i=0}^m i v_i > d$.

(A5) $d \notin \mathcal{M}$.

From Assumption (A2), it is clear that $c(\cdot)$ and $c_u(\cdot) = dc(\cdot)/du$ are nondecreasing. From Assumptions (A4) and (A5), it is easily seen that there is a unique $i_0 \in \mathcal{M}$ such that $0 \leq i_0 < d < i_0 + 1 \leq m$.

Remark 2.1. Assumption (A1) is the usual assumption on the growth rate of the surplus cost function to ensure the existence of solutions to the HJB equation of the optimal control problem. Assumption (A2) is required in order for the optimal control policy to be written explicitly in a simple form. Assumption (A4)—the average capacity of the machine is greater than demand—is necessary in the sense that, if this were not true, then even if the system is always producing at its maximum capacity, it still would not meet the demand and the backlog would build up without bound over time with the consequence that no optimal solution with a finite cost would exist. Assumption (A5) is innocuous and is used for the purpose of ensuring the differentiability of the value function; see Sethi and Zhang (Ref. 1).

In (2), we have defined our cost function to be the limit superior (lim sup) of the finite horizon average costs as the horizon increases, and our optimal control problem to be one of minimizing the cost over the (natural) class of admissible controls. Since the limit of the finite-horizon costs may not exist for all controls in this admissible class, it may be of interest to know the class of controls over which the optimal control also minimizes the limit inferior (lim inf) of the finite-horizon costs. For this purpose, we define a smaller class of controls as follows: A control $u(\cdot) \in \mathcal{A}(k)$ is called stable if it satisfies the condition

$$\lim_{T \rightarrow \infty} E|x(T)|^{\kappa+1}/T = 0, \quad (3)$$

where $x(\cdot)$ is the surplus process corresponding to the control $u(\cdot)$ with $(x(0), \alpha(0)) = (x, k)$ and κ is defined in Assumption (A1). Let $\mathcal{B}(k) \subset \mathcal{A}(k)$ denote the class of stable controls. It can be seen in Section 3 that the set of stable admissible controls $\mathcal{B}(k)$ is nonempty.

We will show in Section 4 that there exists a constant λ^* , independent of the initial condition $(x(0), \alpha(0)) = (x, k)$, and a stable Markov control policy $u^*(\cdot) \in \mathcal{A}(k)$ such that $u^*(\cdot)$ is optimal, i.e., it minimizes the cost

defined by (2) over all $u(\cdot) \in \mathcal{A}(k)$, and furthermore,

$$\lim_{T \rightarrow \infty} (1/T) E \int_0^T (h(x^*(t)) + c(u^*(t))) dt = \lambda^*,$$

where $x^*(\cdot)$ is the surplus process corresponding to $u^*(\cdot)$ with $(x(0), \alpha(0)) = (x, k)$. Moreover, for any other (stable) control $u(\cdot) \in \mathcal{B}(k)$,

$$\liminf_{T \rightarrow \infty} (1/T) E \int_0^T (h(x(t)) + c(u(t))) dt \geq \lambda^*.$$

Since we will use the vanishing discount approach to study our problem, we provide an analysis of the discounted problem in Section 3.

3. Analysis of the Discounted Cost Problem

In order to derive the HJB equation for the average cost control problem formulated above and study the existence of its solutions, we introduce a corresponding control problem with the cost discounted at a rate $\rho > 0$. For $u(\cdot) \in \mathcal{A}(k)$, we define the expected discounted cost as

$$J^\rho(x, k, u(\cdot)) = E \int_0^\infty \exp(-\rho t) (h(x(t)) + c(u(t))) dt.$$

Define the value function of the discounted cost problem as

$$V^\rho(x, k) = \inf_{u(\cdot) \in \mathcal{A}(k)} J^\rho(x, k, u(\cdot)), \quad (4)$$

which we know from Sethi, Soner, Zhang, and Jiang (Ref. 7) to be continuously differentiable in x .

The HJB equation associated with this problem is

$$\rho V^\rho(x, k) = F(k, V_x^\rho(x, k)) + h(x) + QV^\rho(x, \cdot)(k), \quad (5)$$

where $V_x^\rho(\cdot, \cdot)$ is the partial derivative of $V^\rho(\cdot, \cdot)$ with respect to its first variable,

$$F(k, r) = \inf_{0 \leq u \leq k} ((u - d)r + c(u)),$$

and the operator Q is defined by

$$QV^\rho(x, \cdot)(k) = \sum_{k' \neq k} q_{kk'} (V^\rho(x, k') - V^\rho(x, k)). \quad (6)$$

For the reader's convenience, we recapitulate some of the results in Sethi et al. (Ref. 7) concerning the discounted cost problem.

Theorem 3.1. See Sethi et al. (Ref. 7). The value function V^ρ has the following properties:

- (i) $V^\rho(\cdot, k)$ is continuously differentiable and convex for any fixed $k \in \mathcal{M}$. Moreover, there are positive constants C_1, C_2, C_3 such that, for any k ,

$$C_1|x|^{\kappa_0} - C_2 \leq V^\rho(x, k) \leq C_3(1 + |x|^\kappa),$$

where κ, κ_0 are as defined in Assumption (A1).

- (ii) $V^\rho(\cdot, \cdot)$ is the unique solution of the HJB equation (5).

Remark 3.1. Note that the constants C_1, C_2, C_3 may depend on the discount rate ρ .

The optimal control policy is given by

$$u_\rho^*(x, k) = \begin{cases} 0, & V_x^\rho(x, k) > -c_u(0), \\ (c_u)^{-1}(V_x^\rho(x, k)), & -c_u(k) \leq V_x^\rho(x, k) \leq -c_u(0), \\ k, & V_x^\rho(x, k) < -c_u(k), \end{cases} \quad (7)$$

when $c(\cdot)$ is strictly convex, and by

$$u_\rho^*(x, k) = \begin{cases} 0, & V_x^\rho(x, k) > -c, \\ k \wedge d, & V_x^\rho(x, k) = -c, \\ k, & V_x^\rho(x, k) < -c, \end{cases} \quad (8)$$

when $c(u) = cu$ for some constant $c \geq 0$.

In order to study the long-run average cost control problem using the vanishing discount approach, we must first obtain some estimates for the value function $V^\rho(x, k)$.

Lemma 3.1. There exists a constant $C > 0$ such that, for any (x_0, k_0) , $(x, k) \in \mathbb{R} \times \mathcal{M}$, we can find an admissible control $u(\cdot) \in \mathcal{A}(k_0)$ such that, for $r \geq 1$,

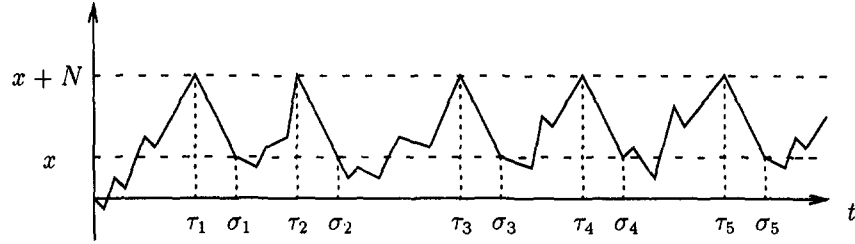
$$E\tau^r \leq C(1 + |x - x_0|^r), \quad (9)$$

where

$$\tau \equiv \inf\{t > 0: (x(t), a(t)) = (x, k)\}, \quad (10)$$

and $x(\cdot)$ is the surplus process corresponding to the control policy $u(\cdot)$ and the initial condition $(x(0), a(0)) = (x_0, k_0)$.

Proof. We provide a proof only in the case where $(x_0, k_0) = (0, 0)$. The proofs in all other cases are similar.

Fig. 1. Surplus process under $u(\cdot)$.

Recall that $\{v_k : k=0, 1, \dots, m\}$ is the stationary distribution of the Markov chain $\alpha(\cdot)$, i.e.,

$$\lim_{t \rightarrow \infty} P(\alpha(t) = k | \alpha(0) = i) = v_k, \quad \forall i \in \mathcal{M}.$$

Notice that, by Assumption (A2), $\alpha(\cdot)$ is a strongly irreducible Markov chain, so we have $v_k > 0$, $k=0, 1, \dots, m$. Since $\mathcal{M}$ is a finite set, we can take $R > 0$ such that, when $t \geq R$,

$$P(\alpha(t) = k | \alpha(0) = i) \geq v_k/2, \quad \forall i, k \in \mathcal{M}. \quad (11)$$

Take N to be a number such that $N > Rd$. We first consider the case where $x + N > 0$. See Fig. 1, plotted for $x > 0$ without loss of generality. In what follows, we construct an appropriate control policy used in the proof. Let

$$u(t) = \alpha(t), \quad x(t) = \int_0^t (u(s) - d) ds, \quad t \leq \tau_1,$$

$$\tau_1 \equiv \inf\{t > 0 : x(t) = x + N\},$$

and

$$u(t) = 0, \quad x(t) = x + N - td, \quad \tau_1 < t \leq \tau_1 + N/d.$$

For convenience in notation, we write

$$\sigma_1 = \tau_1 + N/d.$$

Proceeding in this manner, we can define the required control policy $u(\cdot) \in \mathcal{A}(0)$ inductively,

$$u(t) = \begin{cases} \alpha(t), & \sigma_{n-1} < t \leq \tau_n, \\ 0, & \tau_n < t \leq \sigma_n, \end{cases}$$

where

$$x(t) = \int_0^t (u(s) - d) ds, \quad 0 \leq t \leq \tau_n,$$

$$\tau_n = \inf\{t > \sigma_{n-1}, x(t) = x + N\},$$

$$\sigma_n = \tau_n + N/d.$$

The control policy $u(\cdot)$ can be characterized as follows: Use the maximum available production rate $u(t) = \alpha(t)$ to move the surplus process from 0 or x to $x + N$, and then use zero production rate until the surplus process drops to the level x . A sample path of the surplus process for $x + N > 0$ is shown in Fig. 1.

It is obvious by our construction that $x(\sigma_n) = x$, $n = 1, 2, \dots$. Furthermore, by the strong Markov property of $\alpha(\cdot)$, we have

$$\begin{aligned} P(\tau > \sigma_n) &\leq P(\alpha(\sigma_n) \neq k, \alpha(\sigma_{n-1}) \neq k, \dots, \alpha(\sigma_1) \neq k) \\ &= P(\alpha(\sigma_n) \neq k | \alpha(\sigma_{n-1}) \neq k) \\ &\quad \cdots P(\alpha(\sigma_2) \neq k | \alpha(\sigma_1) \neq k) P(\alpha(\sigma_1) \neq k). \end{aligned} \quad (12)$$

Note that

$$\sigma_i - \sigma_{i-1} \geq N/d > R, \quad i = 1, 2, \dots$$

We set $\sigma_0 = 0$ and $\tau_0 = 0$. By (11), we have

$$P(\alpha(\sigma_1) \neq k) \leq 1 - \delta,$$

$$P(\alpha(\sigma_i) \neq k | \alpha(\sigma_{i-1}) \neq k) \leq 1 - \delta, \quad i = 2, 3, \dots,$$

where

$$\delta \equiv \min\{v_k/2 : k \in \mathcal{M}\}.$$

Then from (12), we have

$$P(\tau > \sigma_n) \leq (1 - \delta)^n. \quad (13)$$

Recall that for $n \geq 2$,

$$\tau_n - \tau_{n-1} = \tau_n - \sigma_{n-1} + N/d$$

and

$$\tau_n - \sigma_{n-1} = \inf\left\{t > 0 : \int_{\sigma_{n-1}}^{\sigma_{n-1}+t} (\alpha(s) - d) ds \geq N\right\}.$$

Apply Lemma 7.1 for any $r \geq 1$ to obtain

$$E|\sigma_n - \sigma_{n-1}|^r \leq C_1(1 + |x + N|^r) \leq C_2(1 + |x|^r), n \geq 1,$$

where $C_1, C_2 > 0$ are constants independent of x . Therefore,

$$\begin{aligned} E\tau^r &= r \int_0^\infty t^{r-1} P(\tau > t) dt \\ &= r \sum_{n=1}^\infty E \int_{\tau_{n-1}}^{\tau_n} t^{r-1} P(\tau > t) dt \\ &\leq r \sum_{n=1}^\infty E \int_{\tau_{n-1}}^{\tau_n} t^{r-1} P(\tau > \tau_{n-1}) dt \\ &\leq r \sum_{n=1}^\infty E \int_{\tau_{n-1}}^{\tau_n} t^{r-1} P(\tau > \sigma_{n-2}) dt \\ &\leq \sum_{n=1}^\infty E(\tau_n - \tau_{n-1})^r (1 - \delta)^{n-2} \\ &\leq C_2(1 + |x|^r) \sum_{n=1}^\infty (1 - \delta)^{n-1} \\ &\leq C(1 + |x|^r). \end{aligned}$$

When $x + N \leq 0$, the proof is the same except that we define $u(t) = 0$ until the time τ_1 , which in this case exactly equals $|x + N|/d$. $\square$

Theorem 3.2. There exists a constant $\rho_0 > 0$ such that $\rho V^\rho(0, 0)$ is bounded for $0 < \rho \leq \rho_0$.

Proof. By Lemma 3.1, we know that there exists a control policy $u(\cdot) \in \mathcal{A}(0)$ such that, for each $r \geq 1$,

$$E\tau_0^r \leq C,$$

where $C > 0$ is a constant which depends on r and

$$\tau_0 = \inf\{t > 0: (x(t), \alpha(t)) = (0, 0)\},$$

with $x(\cdot)$ being the surplus process corresponding to the control $u(\cdot)$ and the initial condition $(x(0), \alpha(0)) = (0, 0)$. By the dynamic programming

principle, we have

$$\begin{aligned} V^\rho(0, 0) &\leq E \left\{ \int_0^{\tau_0} \exp(-\rho t) [h(x(t)) + c(u(t))] dt \right. \\ &\quad \left. + \exp(-\rho \tau_0) V^\rho(x(\tau_0), \alpha(\tau_0)) \right\} \\ &= E \left\{ \int_0^{\tau_0} \exp(-\rho t) [h(x(t)) + c(u(t))] dt \right. \\ &\quad \left. + \exp(-\rho \tau_0) V^\rho(0, 0) \right\}. \end{aligned}$$

Note that

$$|x(t)| \leq (m+d)t, \quad 0 \leq t \leq \tau.$$

Thus by Assumption (A1),

$$\begin{aligned} h(x(t)) &\leq C_1 (1 + |x(t)|^\kappa) \leq C'_1 (1 + t^\kappa), \\ c(u(t)) &\leq c(m), \quad 0 \leq t \leq \tau. \end{aligned}$$

We have

$$\begin{aligned} (1 - E \exp(-\rho \tau_0)) V^\rho(0, 0) &\leq E \int_0^{\tau_0} C_2 (1 + t^\kappa) dt \\ &\leq C'_2 (1 + E \tau_0^{\kappa+1}) \leq C_3. \end{aligned} \quad (14)$$

Now, using the following inequality:

$$1 - \exp(-\rho \tau_0) \geq \rho \tau_0 - \rho^2 \tau_0^2 / 2,$$

we can get

$$(1 - E \exp(-\rho \tau_0)) V^\rho(0, 0) \geq (E \tau_0 - \rho E \tau_0^2 / 2) \cdot \rho V^\rho(0, 0). \quad (15)$$

From the definition of the stopping time τ_0 , we know that $\tau_0 > 0$. Moreover, $E \tau_0$ and $E \tau_0^2$ are finite. Therefore, we have

$$0 < E \tau_0 < \infty, \quad 0 < E \tau_0^2 < \infty.$$

Take

$$\rho_0 = E \tau_0 / E \tau_0^2.$$

By (14) and (15), we have that, for $0 < \rho \leq \rho_0$,

$$\rho V^\rho(0, 0) \leq 2(1 - E \exp(-\rho \tau_0)) V^\rho(0, 0) / E \tau_0 \leq 2C_3 / E \tau_0,$$

which is a constant independent of ρ . $\square$

Let us define the differential discounted value function

$$\bar{V}^\rho(x, k) = V^\rho(x, k) - V^\rho(0, 0), \quad (16)$$

for which the following results can be derived.

Theorem 3.3. The function $\bar{V}^\rho(x, k)$ is convex in x . It is locally uniformly bounded, i.e., there exists a constant $C > 0$ such that

$$|\bar{V}^\rho(x, k)| \leq C(1 + |x|^{\kappa+1}), \quad \forall (x, k) \in \mathbb{R} \times \mathcal{M}, \rho \geq 0. \quad (17)$$

Proof. The convexity of $\bar{V}^\rho(\cdot, k)$ follows from that of $V^\rho(\cdot, k)$. Thus, we need only to show Inequality (17).

We first consider the upper bound for $\bar{V}^\rho(x, k)$. By Lemma 3.1, there exists a constant $C > 0$ and a control $u(\cdot) \in \mathcal{A}(k)$ such that

$$E \tau_0^{\kappa+1} \leq C(1 + |x|^{\kappa+1}),$$

with

$$\tau_0 = \inf\{t > 0 : (x(t), \alpha(t)) = (0, 0)\}, \quad (18)$$

where $x(\cdot)$ is the surplus process corresponding to $u(\cdot)$ and the initial condition $(x(0), \alpha(0)) = (x, k)$. Then, from the dynamic programming principle, we have

$$\begin{aligned} V^\rho(x, k) &\leq E \left\{ \int_0^{\tau_0} \exp(-\rho t) [h(x(t)) + c(u(t))] dt \right. \\ &\quad \left. + \exp(-\rho \tau_0) V(x(\tau_0), \alpha(\tau_0)) \right\} \\ &= E \left\{ \int_0^{\tau_0} \exp(-\rho t) [h(x(t)) + c(u(t))] dt + \exp(-\rho \tau_0) V(0, 0) \right\} \\ &\leq E \left\{ \int_0^{\tau_0} \exp(-\rho t) [h(x(t)) + c(u(t))] dt + V(0, 0) \right\}. \end{aligned} \quad (19)$$

Note that

$$|x(t)| \leq |x| + (m + d)t, \quad 0 \leq t \leq \tau_0.$$

Thus by Assumption (A1),

$$\begin{aligned} h(x(t)) &\leq C_1(1 + |x|^\kappa + t^\kappa), \\ c(u(t)) &\leq c(m), \quad 0 \leq t \leq \tau_0. \end{aligned}$$

Therefore, by (19) and (18) we have

$$\begin{aligned} \bar{V}^\rho(x, k) &= V^\rho(x, k) - V^\rho(0, 0) \\ &\leq E \int_0^{\tau_0} C_2(1 + |x|^\kappa + t^\kappa) dt \\ &\leq C_2'(1 + |x|^\kappa E\tau_0 + E\tau_0^{\kappa+1}) \\ &\leq C(1 + |x|^{\kappa+1}). \end{aligned} \quad (20)$$

Next, we consider the lower bound of $\bar{V}^\rho(x, k)$. By Lemma 3.1, there exists an admissible control $u(\cdot) \in \mathcal{A}(0)$ such that

$$E\tau^{\kappa+1} \leq C(1 + |x|^{\kappa+1}), \quad \forall x \in \mathbb{R}, \quad (21)$$

where

$$\tau \equiv \inf\{t > 0: (x(t), \alpha(t)) = (x, k)\},$$

$C > 0$ is a constant independent of x , and $x(\cdot)$ is the surplus process corresponding to the control policy $u(\cdot)$ and the initial condition $(x(0), \alpha(0)) = (0, 0)$. Apply the dynamic programming principle to obtain

$$\begin{aligned} V^\rho(0, 0) &\leq E \left\{ \int_0^\tau \exp(-\rho t) [hg(x(t)) + c(u(t))] dt \right. \\ &\quad \left. + \exp(-\rho \tau) V^\rho(x(\tau), \alpha(\tau)) \right\} \\ &= E \left\{ \int_0^\tau \exp(-\rho t) [h(x(t)) + c(u(t))] dt + \exp(-\rho \tau) V^\rho(x, k) \right\} \\ &\leq E \left\{ \int_0^\tau \exp(-\rho t) [h(x(t)) + c(u(t))] dt + V^\rho(x, k) \right\}. \end{aligned}$$

Therefore, by noting that

$$x(t) \leq (m + d)t$$

and using Assumptions (A1) and (A2), we have

$$\begin{aligned}
 \bar{V}^\rho(x, k) &= V^\rho(x, k) - V^\rho(0, 0) \\
 &\geq E \left\{ - \int_0^\tau \exp(-\rho t) [h(x(t)) + c(u(t))] dt \right. \\
 &\quad \left. + (1 - \exp(-\rho \tau)) V^\rho(x, k) \right\} \\
 &\geq E \left\{ - \int_0^\tau \exp(-\rho t) [h(x(t)) + c(u(t))] dt \right\} \\
 &\geq -E \left\{ \int_0^\tau C_1(1 + t^\kappa) dt \right\} \\
 &\geq -C_2(1 + E\tau^{\kappa+1}) \\
 &\geq -C(1 + |x|^{\kappa+1}). \tag{22}
 \end{aligned}$$

The theorem is thus proved by combining (20) and (22). $\square$

Corollary 3.1. $\bar{V}^\rho(x, k)$ is locally uniformly Lipschitz continuous in x with respect to $\rho > 0$; i.e., for any $X > 0$, there exists a constant $C > 0$, independent of ρ , such that

$$|\bar{V}^\rho(x, k) - \bar{V}^\rho(x', k)| \leq C|x - x'|,$$

for all $k \in \mathcal{M}$ and all $|x| \leq X$ and $|x'| \leq X$.

Proof. The proof follows immediately from Theorem 3.3 and a result in Rockafellar (Ref. 10, Chapter 2, Theorem 10.6). For the reader's convenience, a proof is also provided in Section 7. $\square$

Corollary 3.2. For $(x, k) \in \mathbb{R} \times \mathcal{M}$, there is a subsequence of ρ , still denoted by ρ , such that

$$\lambda^* = \lim_{\rho \rightarrow 0} \rho V^\rho(x, k), \quad V(x, k) = \lim_{\rho \rightarrow 0} \bar{V}^\rho(x, k). \tag{23}$$

Moreover, the convergence is locally uniform in (x, k) and $V(\cdot)$ is locally Lipschitz continuous.

Proof. The proof follows immediately from Theorem 3.2, Theorem 3.3, Corollary 3.1, and the Arzela–Ascoli theorem $\square$

4. Verification Theorem

The HJB equation associated with the long-run average cost optimal control problem formulated in Section 2 takes the following form:

$$\lambda = F(k, W_x(x, k)) + h(x) + QW(x, \cdot)(k), \quad (24)$$

where λ is a constant and W is a real-valued function defined on $\mathbb{R} \times \mathcal{M}$.

Before we define a solution to the HJB equation (24), we first introduce some notation. Let $\mathcal{G}$ denote the family of real-valued functions $W(\cdot, \cdot)$ defined on $\mathbb{R} \times \mathcal{M}$ such that:

- (i) $W(\cdot, k)$ is convex;
- (ii) $W(\cdot, k)$ is continuously differentiable;
- (iii) $W(\cdot, k)$ has polynomial growth; i.e., there are constants $l, C > 0$ such that

$$|W(x, k)| \leq C(1 + |x|^l), \quad \forall x \in \mathbb{R}.$$

A solution to the HJB equation (24) is a pair (λ, W) with λ a constant and $W \in \mathcal{G}$. The function W is called a potential function for the control problem, if λ is the minimum long-run average cost.

We now show that the limit (λ^*, V) defined by (23) in Corollary 3.2 is indeed a solution to the HJB equation (24). Since the HJB equation involves the partial derivative $V_x(\cdot, \cdot)$, which may not exist for the defined function $V(\cdot, \cdot)$, we must first interpret the solution of the HJB equation (24) in the viscosity sense. This is done in Theorem 4.1. In Theorem 4.2, we show that $V(\cdot, k)$ is continuously differentiable, and therefore (λ^*, V) is indeed a classical solution. We refer to Fleming and Soner (Ref. 8) for the definition of viscosity solutions and some relevant results.

Theorem 4.1. (λ^*, V) is a viscosity solution to the HJB equation (24). Moreover, the constant λ^* is unique in the following sense: if (λ, V') is another viscosity solution to (24), then $\lambda = \lambda^*$.

Proof. By Corollary 3.2, we know that the convergence in (23) is locally uniform in (x, k) , where $\bar{V}^\rho$ is defined by (16). From Sethi and Zhang (Ref. 1), we know that V^ρ is the classical solution, and thus a viscosity solution to (5). As a result, $\bar{V}^\rho$ is a viscosity solution to

$$\rho \bar{V}^\rho(x, k) + \rho V^\rho(0, 0) = F(k, V_x^\rho(x, k)) + Q\bar{V}^\rho(x, \cdot)(k). \quad (25)$$

In (25), $\rho V^\rho(0, 0) \rightarrow \lambda^*$ and $\rho \bar{V}^\rho(x, k) \rightarrow 0$ locally uniformly. Using the properties of viscosity solutions, we can conclude that (λ^*, V) is a solution to (24). The fact that

$$\lambda^* = \lim_{\rho \rightarrow 0} \rho V^\rho(x, k), \quad \text{for any } (x, k) \in \mathbb{R} \times \mathcal{M},$$

is easy to see from the facts that

$$\rho V^\rho(0, 0) \rightarrow \lambda^*, \quad \rho \bar{V}^\rho(x, k) \rightarrow 0,$$

and

$$V^\rho(x, k) = \bar{V}^\rho(x, k) + V^\rho(0, 0).$$

The uniqueness of λ^* can be shown similarly as in Sethi and Zhang (Ref. 1, Appendix G, Theorem G.1). $\square$

Remark 4.1. While the proof of Sethi and Zhang (Ref. 1, Appendix G, Theorem G.1) works for the uniqueness of λ^* , it cannot be adapted to prove whether V and V' are equal. On the other hand, we do not need V to be unique for the purpose of this paper.

In the next theorem, we derive the smoothness property of the function $V(x, k)$ and a bound for it.

Theorem 4.2. The function $V(x, k)$ is continuously differentiable in x , and (λ^*, V) is a classical solution to the HJB equation. Moreover, $V(x, k)$ is convex in x and

$$|V(x, k)| \leq C(1 + |x|^{\kappa+1}).$$

Proof. Since the function $V(x, k)$ is convex in x , it suffices to show that the subdifferential $D^-V(x, k)$ is a singleton in view of Lemma 7.2 in the Appendix (Section 7). Note that the map

$$r \rightarrow F(k, r) := \inf_{0 \leq u \leq k} \{(u - d)r + c(u)\}$$

is not constant on any nontrivial interval. It follows from the proof of Theorem 3.1 in Chapter 3 of Sethi and Zhang (Ref. 1) that

$$\lambda^* = F(k, r) + h(x) + QV(x, \cdot)(k), \quad \forall r \in D^-V(x, k).$$

Therefore, $F(k, r)$ is a constant on the convex set $D^-V(x, k)$, which implies that $D^-V(x, k)$ is a singleton. The proof of the convexity of $V(x, k)$ follows from the convexity of $\bar{V}^\rho(x, k)$. The upper bound on $V(x, k)$ follows from Theorem 3.3. $\square$

The following verification theorem can be proved now.

Theorem 4.3. Let (λ, W) be a solution to the HJB equation (24). Then:

- (i) if there is a control $u^*(\cdot) \in \mathcal{A}(k)$ such that

$$\begin{aligned} F(\alpha(t), W_x(x^*(t), \alpha(t))) \\ = (u^*(t) - d)W_x(x^*(t), \alpha(t)) + c(u^*(t)), \end{aligned} \quad (26)$$

for a.e. $t \geq 0$ with probability 1, where $x^*(\cdot)$ is the surplus process corresponding to the control $u^*(\cdot)$, and

$$\lim_{T \rightarrow \infty} W(x^*(T), \alpha(T))/T = 0, \quad (27)$$

then

$$\lambda = J(x, k, u^*(\cdot)).$$

- (ii) For any $u(\cdot) \in \mathcal{A}(k)$, we have $\lambda \leq J(x, k, u(\cdot))$, i.e.,

$$\limsup_{T \rightarrow \infty} E \int_0^T (h(x(t)) + c(u(t))) dt \geq \lambda.$$

- (iii) Furthermore, for any (stable) control policy $u(\cdot) \in \mathcal{B}(k)$, we have

$$\liminf_{T \rightarrow \infty} (1/T) E \int_0^T (h(x(t)) + c(u(t))) dt \geq \lambda. \quad (28)$$

Proof. We begin with the proof of part (i). Since (λ, W) is a solution to the HJB equation (24) and $u^*(\cdot)$ satisfies the condition (26), we have

$$\begin{aligned} (u^*(t) - d)W_x(x^*(t), \alpha(t)) + QW(x^*(t), \cdot)(\alpha(t)) \\ = \lambda - h(x^*(t)) - c(u^*(t)). \end{aligned} \quad (29)$$

Since $W \in \mathcal{G}$, we can apply the Dynkin formula [see Fleming and Rishel (Ref. 9)] and (29) to get

$$\begin{aligned} EW(x^*(T), \alpha(T)) \\ = W(x, k) + E \int_0^T ((u^*(t) - d)W_x(x^*(t), \alpha(t)) \\ + QW(x^*(t), \cdot)(\alpha(t))) dt \\ = W(x, k) + E \int_0^T (\lambda - h(x^*(t)) - c(u^*(t))) dt \\ = W(x, k) + \lambda T - E \int_0^T (h(x^*(t)) + c(u^*(t))) dt. \end{aligned} \quad (30)$$

We can rewrite (30) as

$$\begin{aligned}\lambda &= (1/T)(EW(x^*(T), \alpha(T)) - W(x, k)) \\ &\quad + (1/T)E \int_0^T (h(x^*(t)) + c(u^*(t))) dt,\end{aligned}$$

and the first part of the theorem is proved by taking the limit as $T \rightarrow \infty$ and using the condition (27).

For the proof of part (iii), if $u(\cdot) \in \mathcal{B}(k)$, then by Theorem 4.2 we know that

$$\lim_{T \rightarrow \infty} W(x(T), \alpha(T))/T = 0.$$

Moreover, from the HJB equation (24), we have

$$\begin{aligned}(u(t) - d)W_x(x(t), \alpha(t)) + QW(x(t), \cdot)(\alpha(t)) \\ \geq \lambda - h(x(t)) - c(u(t)).\end{aligned}$$

Now, (28) can be proved similarly as before.

Finally, we apply Theorem 7.1 (Tauberian theorem) in the Appendix (Section 7) to show part (ii), i.e., the optimality of the control $u^*(\cdot)$ in the (natural) class of all admissible controls. Let $u(\cdot) \in \mathcal{A}(k)$ be any policy, and let $x(\cdot)$ be the corresponding surplus process. Suppose that

$$J(x, k, u(\cdot)) < \lambda. \quad (31)$$

Set

$$f(t) = E(h(x(t)) + c(u(t))).$$

Without loss of generality, we may assume that

$$\int_0^t f(s) ds < \infty,$$

for each $t > 0$, otherwise $J(x, k, u(\cdot)) = \infty$. Note that

$$J(x, k, u(\cdot)) = \limsup_{t \rightarrow \infty} (1/t) \int_0^t f(s) ds,$$

while

$$\rho J^\rho(x, k, u(\cdot)) = \rho \int_0^\infty \exp(-\rho s) f(s) ds.$$

Therefore, we can apply Theorem 7.1 in the Appendix (Section 7) to obtain

$$\limsup_{\rho \rightarrow 0} \rho J^\rho(x, k, u(\cdot)) < \lambda. \quad (32)$$

On the other hand, we know from Theorem 4.1 that

$$\lim_{\rho \rightarrow 0} \rho V^\rho(x, k) = \lambda^* = \lambda.$$

This equation and (32) imply the existence of a $\rho > 0$ such that

$$\rho J^\rho(x, k, u(\cdot)) < \rho V^\rho(x, k),$$

which contradicts the definition of $V^\rho(x, k)$. Thus, (ii) is proved. $\square$

Remark 4.2. In the simple special case solved by Bielecki and Kumar (Ref. 2), we note that they minimize the liminf of the long-run average cost over the class of admissible stable controls. They do not consider the minimization of limsup defined in (2). That is, they prove (i) and (iii), but not (ii).

5. Existence and Characterization of Optimal Control

From Theorem 4.2, we know that the function $V \in \mathcal{G}$, and that it is also a solution of the HJB equation (24). The function V is sometimes referred to as the relative value function. In a way similar to (7) and (8), let us now define a control policy $u^*(\cdot, \cdot)$ via the relative value function $V(\cdot, \cdot)$ as follows:

$$u^*(x, k) = \begin{cases} 0, & V_x(x, k) > -c_u(0), \\ (c_u)^{-1}(-V_x(x, k)), & -c_u(k) \leq V_x(x, k) \leq -c_u(0), \\ k, & V_x(x, k) < -c_u(k), \end{cases} \quad (33)$$

if the function $c(\cdot)$ is strictly convex, or

$$u^*(x, k) = \begin{cases} 0, & V_x(x, k) > -c, \\ k \wedge d, & V_x(x, k) = -c, \\ k, & V_x(x, k) < -c, \end{cases} \quad (34)$$

if $c(u) = cu$. Therefore, the control policy $u^*(\cdot, \cdot)$ satisfies the condition (26).

Next, we devote ourselves to proving that $u^*(\cdot, \cdot)$ is a stable control. For this, we first derive some intermediate results.

Lemma 5.1. For each $k \in \mathcal{M}$, we have

$$\inf_{x \in \mathbb{R}} V(x, k) > -\infty. \quad (35)$$

Proof. Let (x_ρ, k_ρ) be the minimum point of the value function $V^\rho(\cdot, \cdot)$. Then, we can write

$$\begin{aligned} \bar{V}^\rho(x, k) &= V^\rho(x, k) - V^\rho(x_\rho, k_\rho) + V^\rho(x_\rho, k_\rho) - V^\rho(0, 0) \\ &\geq V^\rho(x_\rho, k_\rho) - V^\rho(0, 0). \end{aligned}$$

By Lemma 3.3, we have

$$\bar{V}^\rho(x, k) \geq -C(1 + |x_\rho|^\kappa).$$

Since $\bar{V}^\rho(x, k) \rightarrow V(x, k)$, we need only to show that $\{x_\rho\}$ is bounded.

From the definition of (x_ρ, k_ρ) , we can see that

$$V_x^\rho(x_\rho, k_\rho) = 0, \quad \forall \rho > 0.$$

Recall that $V^\rho(\cdot, \cdot)$ is a solution of the HJB equation (5). Thus,

$$\begin{aligned} \rho V^\rho(x_\rho, k_\rho) &= \inf_{0 \leq u \leq k_\rho} \{(u - d)V_x^\rho(x_\rho, k_\rho) + c(u)\} \\ &\quad + QV^\rho(x_\rho, \cdot)(k_\rho). \end{aligned}$$

From the fact that (x_ρ, k_ρ) is the minimum point of $V^\rho(\cdot, \cdot)$, we can conclude that $QV^\rho(x_\rho, \cdot)(k_\rho) \geq 0$. Therefore,

$$\rho V^\rho(x_\rho, k_\rho) \geq h(x_\rho).$$

From Lemma 3.2, we know that there exists a constant $C > 0$ such that

$$\rho V^\rho(x_\rho, k_\rho) \leq \rho V^\rho(0, 0) \leq C, \quad \forall \rho > 0.$$

Therefore,

$$h(x_\rho) \leq C,$$

and the boundedness of $\{x_\rho\}$ follows from Assumption (A1). $\square$

In order to state and prove the next lemma, define

$$U(k) = \{x: V_x(x, k) > -c_u(0)\},$$

$$L(k) = \{x: V_x(x, k) < -c_u(k)\}.$$

Lemma 5.2. The sets $U(k)$ and $L(k)$ are nonempty for each $k \in \mathcal{M}$.

Proof. Define

$$U^0(k) = \{x: V_x(x, k) > 0\}.$$

By Assumption (A1), we know that $-c_u(0) \leq 0$. Since $V_x(\cdot, k)$ is nondecreasing, we have $U^0(k) \subset U(k)$. Thus, in order to prove that $U(k) \neq \emptyset$, it suffices to show that $U^0(k) \neq \emptyset$. If $U^0(k) = \emptyset$, we will have

$$V_x(x, k) \leq 0, \quad \forall x \in \mathbb{R}. \quad (36)$$

Using the fact that $V(\cdot, k)$ is a convex function bounded from below, we can conclude that

$$V_x(x, k) \rightarrow 0, \quad \text{as } x \rightarrow \infty.$$

Thus, we have

$$F(k, V_x(x, k)) = \inf_{0 \leq u \leq k} \{(u-d)V_x(x, k) + c(u)\} \rightarrow 0, \quad \text{as } x \rightarrow \infty.$$

Since $V(\cdot, \cdot)$ is a solution of the HJB equation (24) and since $h(x) \rightarrow \infty$ as $x \rightarrow \infty$, we can see that

$$QV(x, \cdot)(k) \rightarrow -\infty, \quad \text{as } x \rightarrow \infty. \quad (37)$$

Note from (36) that $V(\cdot, k)$ is decreasing. By definition, we know that

$$QV(x, \cdot)(k) = \sum_{k' \neq k} q_{kk'}(V(x, k') - V(x, k)).$$

Moreover, from Assumption (A2), specifying that the generator Q is strongly irreducible, there is a $k' \neq k$ such that $q_{kk'} > 0$. Then, (37) leads to

$$V(x, k') \rightarrow -\infty,$$

which is a contradiction to (35). Therefore, we have proved that

$$U(k) \supset U^0(k) \neq \emptyset.$$

Similarly, we can show that $L(k) \neq \emptyset$. If $L(k) = \emptyset$, then

$$V_x(x, k) \geq -c_u(k), \quad \forall x \in \mathbb{R},$$

and thus $F(x, k, V_x(x, k))$ is bounded from below for $x \rightarrow -\infty$. By letting $x \rightarrow -\infty$ and noting that $h(x) \rightarrow \infty$, we can get a contradiction as above. $\square$

From the convexity of the function $V(\cdot, k)$, there are x_k, y_k , $-\infty < y_k < x_k < \infty$, such that

$$U(x) = (x_k, \infty), \quad L(k) = (-\infty, y_k).$$

The control policy $u^*(\cdot, \cdot)$ can be written as

$$u^*(x, k) = \begin{cases} 0, & x > x_k, \\ (c_u)^{-1}(-V_x(x, k)), & y_k \leq x \leq x_k, \\ k, & x < y_k. \end{cases}$$

Theorem 5.1. The control policy $u^*(\cdot, \cdot)$ is stable.

Proof. Let $x^*(\cdot)$ denote the surplus process corresponding to $u^*(\cdot, \cdot)$ with $x(0) = x$ and $\alpha(0) = i$; see Reference 1, Appendix J. Let

$$N = \max\{|x_k|, |y_k| : k \in \mathcal{M}\}.$$

Then, $u^*(\cdot, \cdot)$ has the following property:

$$u^*(x, k) = \begin{cases} 0, & x \geq N, \\ k, & x \leq -N. \end{cases}$$

Let $x^*(\cdot)$, with $x(0) = x$ and $\alpha(0) = i$, be the surplus process corresponding to the control policy defined by

$$u(y, k) = \begin{cases} 0, & y > 0, \\ k \wedge d, & y = 0, \\ k, & y < 0. \end{cases}$$

It is easy to see that

$$x^*(t) \leq x(t) \leq x^*(t) + 2N,$$

and thus for any $l > 0$, in view of the proof of Lemma 3.2, we have

$$E|x^*(t)|^l \leq C_1 E|x(t)|^l + C_2(2N)^l,$$

where $C_1, C_2 > 0$ are constants. Take $l = \kappa + 1$ to conclude that

$$\lim_{t \rightarrow \infty} E|x^*(t)|^{\kappa+1}/t = 0.$$

□

Now, we are in the position to state and prove the following theorem.

Theorem 5.2. The control policy $u^*(\cdot, \cdot)$, defined in (33) or (34), as the case may be, is optimal.

Proof. By Theorem 4.3, we need only to show that

$$\lim_{t \rightarrow \infty} V(x^*(t), a(t))/t = 0.$$

But this is implied by Theorem 4.2 and the fact that $u^*(\cdot, \cdot)$ is a stable control (Theorem 5.1). $\square$

Remark 5.1. When $c(u) = 0$, i.e., there is no product cost in the model, the optimal control policy can be chosen to be the so-called hedging point policy, which has the following form: there are real numbers x_k , $k = 1, \dots, m$, such that

$$u^*(x, k) = \begin{cases} 0, & x > x_k, \\ k \wedge d, & x = x_k, \\ k, & x < x_k. \end{cases}$$

6. Conclusions

In this paper, we have used the vanishing discount approach to develop the theory of dynamic programming for single or parallel machines, convex cost stochastic manufacturing systems, with the long-run average cost minimization criterion. A flowchart of our approach is presented in Fig. 2. We have generalized previous results as well as provided a theoretical framework for various heuristic analyses of such systems in the literature. A multi-product extension of this paper is currently in progress.

Further research should focus on extending the analysis to more complex manufacturing systems such as flowshops and jobshops. For such models with discounted cost criteria, see Sethi and Zhang (Ref. 1).

7. Appendix

In this section, we derive auxiliary results used earlier in the paper.

Lemma 7.1. Let $\sigma = \inf\{t: \int_0^t (a(s) - d) ds = l\}$, for any $l > 0$. Then, for any $r \geq 1$, there is a constant C independent of l such that $E\sigma^r \leq C(l^r + 1)$.

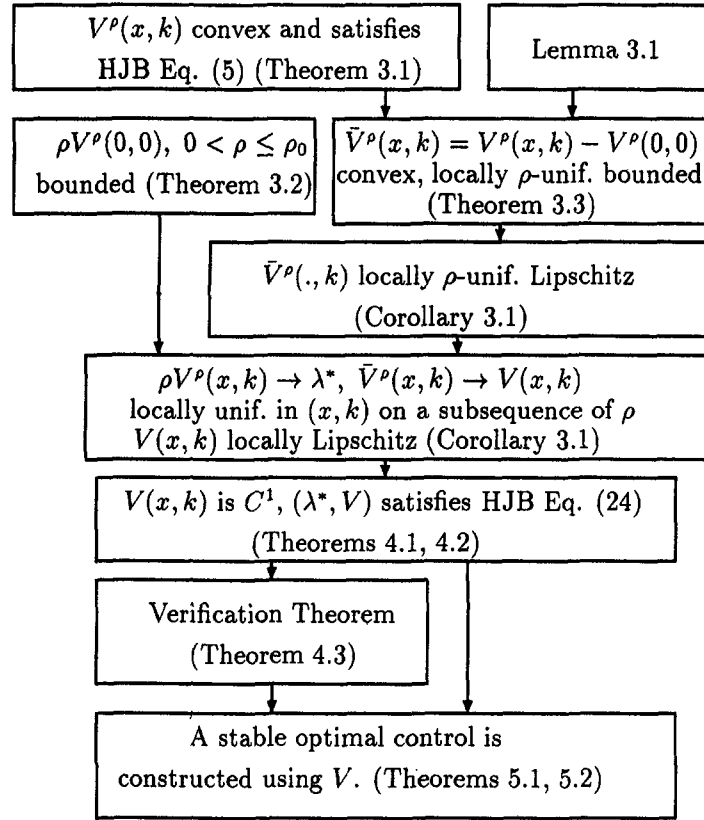


Fig. 2. Flowchart of the vanishing discount approach.

Proof. It can be shown, as in Appendix C of Sethi and Zhang (Ref. 1), that there exists a constant $C' > 0$ such that, for any $t > 0$,

$$E \exp\left((1/\sqrt{t+1}) \left| \int_0^t (a(s) - \bar{a}) ds \right| \right) \leq C', \quad (38)$$

where $\bar{a}$ denotes the average capacity as defined in Section 2.

Note by the definition of σ that

$$\begin{aligned} P(\sigma > t) &\leq P\left(\int_0^t (a(s) - d) ds < l\right) \\ &= P\left(\int_0^t (a(s) - \bar{a}) ds + (\bar{a} - d)t < l\right). \end{aligned}$$

Since $\bar{a} > d$ by Assumption (A4), we have for $t \geq l/(\bar{a} - d)$,

$$\begin{aligned} P(\sigma > t) &\leq P\left(\left|\int_0^t (\alpha(s) - d) ds\right| > t(\bar{a} - d) - l\right) \\ &= P\left((1/\sqrt{t+1})\left|\int_0^t (\alpha(s) - d) ds\right| > (t(\bar{a} - d) - l)/\sqrt{t+1}\right) \\ &\leq \exp\left(-(t(\bar{a} - d) - l)/\sqrt{t+1}\right) \\ &\quad \times E \exp\left((1/\sqrt{t+1})\left|\int_0^t (\alpha(s) - d) ds\right|\right). \end{aligned}$$

In view of this and Inequality (38), we have

$$\begin{aligned} P(\sigma \geq t) &\leq C' \exp(-t(\bar{a} - d)/\sqrt{t+1} + l/\sqrt{t+1}) \\ &= C' \exp\{-t(\bar{a} - d)/2\sqrt{t+1} + (-t(\bar{a} - d)/2\sqrt{t+1} + l/\sqrt{t+1})\} \\ &\leq C' \exp(-t(\bar{a} - d)/2\sqrt{t+1}), \quad t \geq 2l/(\bar{a} - d). \end{aligned}$$

Therefore,

$$\begin{aligned} E\sigma^r &= r \int_0^\infty t^{r-1} P(\sigma \geq t) dt \\ &\leq r \int_0^{2l/(\bar{a} - d)} t^{r-1} dt + rC' \int_{2l/(\bar{a} - d)}^\infty t^{r-1} \exp(-t(\bar{a} - d)/2\sqrt{t+1}) dt \\ &\leq C(l^r + 1), \end{aligned}$$

for some constant $C > 0$, which is independent of l . $\square$

Next, we prove the following Tauberian theorem which is used in Section 4.

Theorem 7.1. If $f(t)$ is a nonnegative Borel measurable function defined on $[0, \infty)$, then

$$\limsup_{\rho \rightarrow 0} \rho \int_0^\infty \exp(-\rho t) f(t) dt \leq \limsup_{t \rightarrow \infty} (1/t) \int_0^t f(s) ds. \quad (39)$$

Proof. We write

$$\limsup_{t \rightarrow \infty} (1/t) \int_0^t f(s) ds = c.$$

Without loss of generality, we may assume that $c < \infty$. For each $\epsilon > 0$, there exists M such that, for each $s > M$,

$$\int_0^s f(t) dt \leq (c + \epsilon)s.$$

Then,

$$\begin{aligned} & \rho \int_0^\infty \exp(-\rho t) f(t) dt \\ &= \int_0^\infty f(t) \int_t^\infty \rho^2 \exp(-\rho s) ds dt \\ &= \int_0^\infty \rho^2 \exp(-\rho s) \int_0^s f(t) dt ds \\ &= \int_M^\infty \rho^2 \exp(-\rho s) \int_0^s f(t) dt ds + \int_0^M \rho^2 \exp(-\rho s) \int_0^s f(t) dt ds \\ &\leq \int_M^\infty \rho^2 \exp(-\rho s) (c + \epsilon)s ds + \int_0^M \rho^2 \exp(-\rho s) \int_0^s f(t) dt ds \\ &\leq \int_0^\infty \rho^2 \exp(-\rho s) (c + \epsilon)s ds + \int_0^M \rho^2 \exp(-\rho s) \int_0^s f(t) dt ds \\ &= (c + \epsilon) + \int_0^M \rho^2 \exp(-\rho s) \int_0^s f(t) dt ds. \end{aligned} \quad (40)$$

Obviously, the second term on the right-hand side of (40) goes to 0 as $\rho \rightarrow 0$. Therefore, the left-hand side of (40) does not exceed $c + \epsilon$. Since ϵ can be arbitrarily small, we have Inequality (39). $\square$

In what follows, we give some results concerning convex functions. We begin with some definitions. For any function $f(\cdot) : \mathbb{R} \mapsto \mathbb{R}$, the superdifferential $D^+f(x)$ and the subdifferential $D^-f(x)$ of the function f at x are

defined, respectively, as follows:

$$D^+f(x) = \left\{ r \in \mathbb{R} : \limsup_{h \rightarrow 0} [f(x+h) - f(x) - hr] / |r| \leq 0 \right\},$$

$$D^-f(x) = \left\{ r \in \mathbb{R} : \liminf_{h \rightarrow 0} [f(x+h) - f(x) - hr] / |r| \geq 0 \right\}.$$

The results stated in the following lemma can be found in Clarke (Ref. 11).

Lemma 7.2. For a convex function $f: \mathbb{R} \mapsto \mathbb{R}$, we have the following properties:

- (i) The subdifferential $D^-f(x)$ is always nonempty. The superdifferential $D^+f(x)$ is empty unless f is differentiable at x .
- (ii) f is differentiable at x if and only if $D^-f(x)$ is a singleton, in which case, $D^-f(x) = D^+f(x)$.
- (iii) If the function f is differentiable on $\mathbb{R}$, then it is continuously differentiable on $\mathbb{R}$.

Lemma 7.3. The function $\bar{V}^\rho(x, k)$, $\rho > 0$, defined by the relation (16), are locally uniformly Lipschitz continuous in x . That is, for any bounded interval $I \subset \mathbb{R}$, there exists a constant $C > 0$ such that

$$|\bar{V}^\rho(x, k) - \bar{V}^\rho(x', k)| \leq C|x - x'|,$$

for all $x, x' \in I$, $\rho > 0$.

Proof. We will prove a more general result for convex functions. If the family of functions $\{f_\rho(\cdot)\}$ defined on a convex set $D \subset \mathbb{R}^n$ is locally bounded, then it is locally uniformly Lipschitz continuous.

Let B and B' , $B \subset B' \subset \mathbb{R}^n$ be balls of radii r and $r+1$, respectively. For $x, x' \in B \cap D$ and $0 \leq \lambda \leq 1$, we have by the definition of convex functions,

$$\begin{aligned} f_\rho(x) - f_\rho(x') &= f(\lambda x' + (1-\lambda)(x - \lambda x') / (1-\lambda)) - f_\rho(x') \\ &\leq \lambda f_\rho(x') - (1-\lambda) f_\rho((x - \lambda x') / (1-\lambda)) - f_\rho(x') \\ &= (1-\lambda) [f_\rho((x - \lambda x') / (1-\lambda)) - f_\rho(x')]. \end{aligned} \quad (41)$$

Without loss of generality, we may take

$$|x - x'| \leq 1.$$

Let

$$\lambda = 1 - |x - x'|.$$

We can then write (41) as

$$\begin{aligned} f_{\rho}(x) - f_{\rho}(x') &\leq |x - x'| [f_{\rho}(x' + (x - x')/|x - x'|) - f_{\rho}(x')] \\ &\leq |x - x'| \cdot 2 \sup_{y \in B'} |f_{\rho}(y)|. \end{aligned}$$

Now, the conclusion follows from the fact that the functions $f_{\rho}(\cdot)$, $\rho > 0$, are locally uniformly bounded. $\square$

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